

Modeling acoustic scattering by various geometries

Vaibhav Chhaya

Project goals

Develop a modeling framework to compute the target strength and other acoustic scattering metrics from discrete biological structures.

- 1) Validate finite-element modeling protocol against analytical scattering benchmarks.
- 2) Apply the validated protocol to CT-derived models of fish.

Benchmarking models



**Chu's analytical
models for discrete
scatterers**

Partial wave
expansion

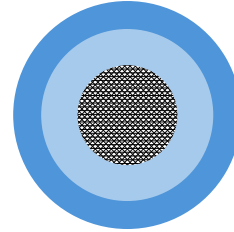
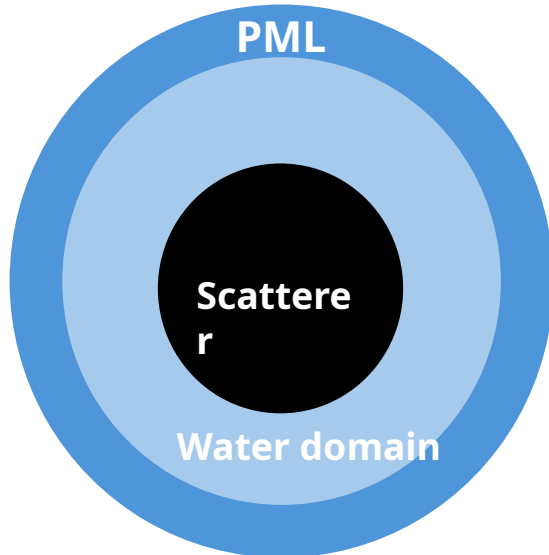
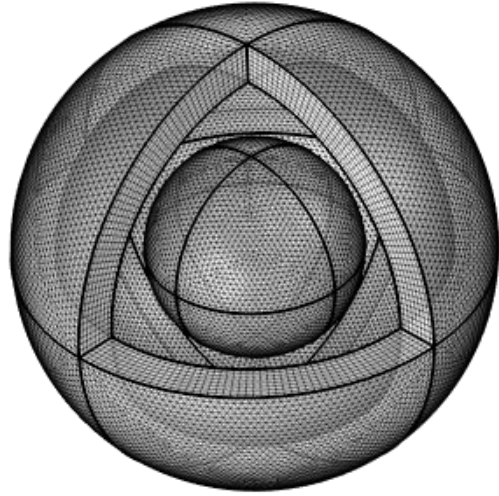
VS



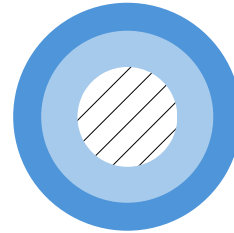
**COMSOL
Multiphysics
numerical models**

Finite element
analysis

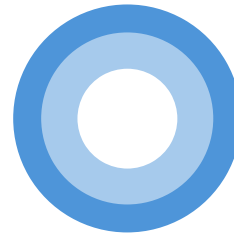
Types of sphere scattering models



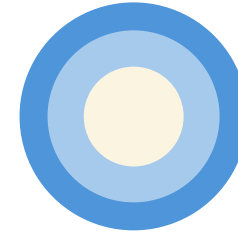
Rigid sphere



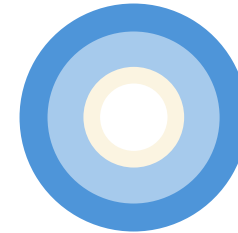
Soft sphere



Air Bubble

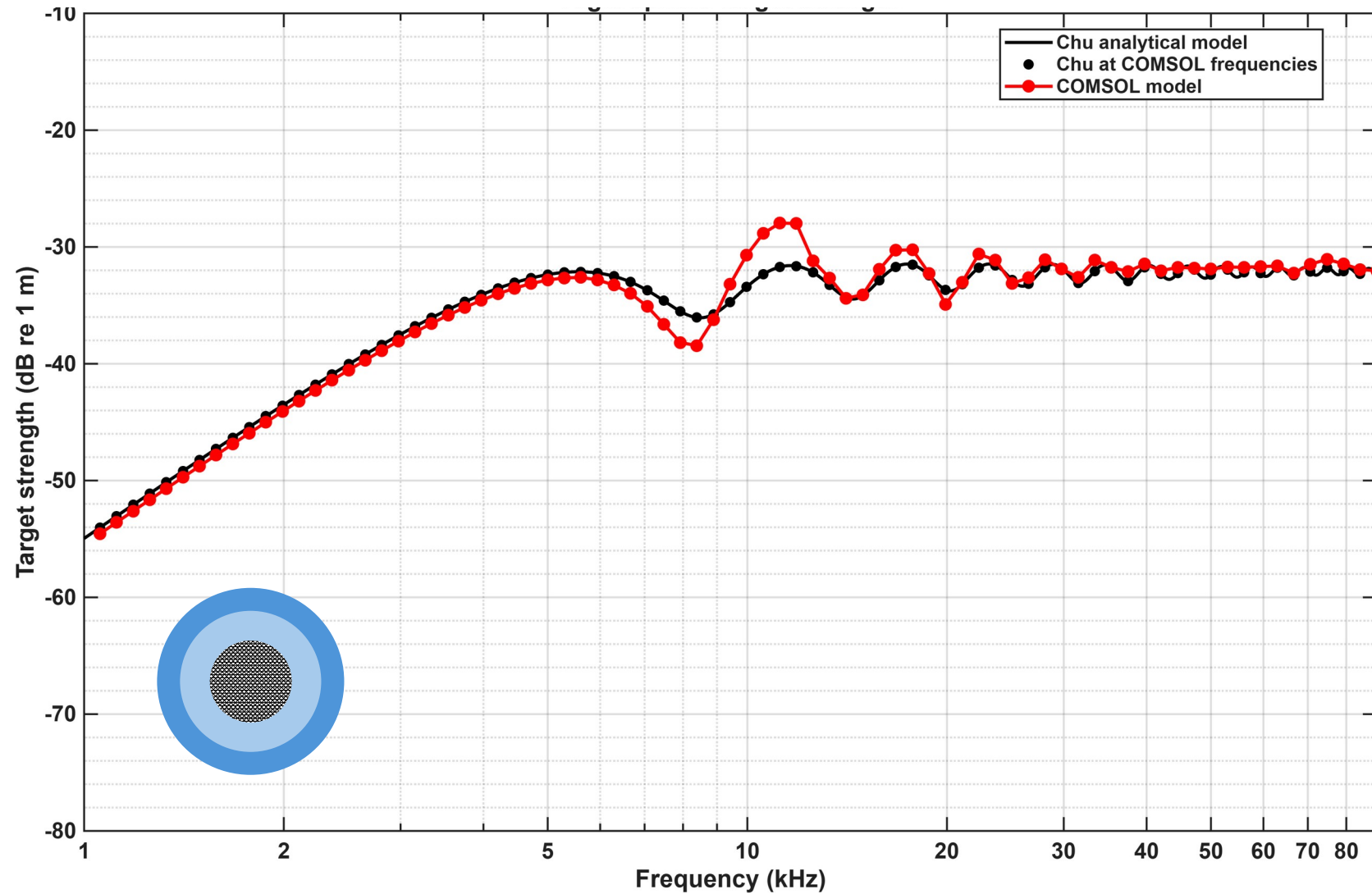


Bone sphere

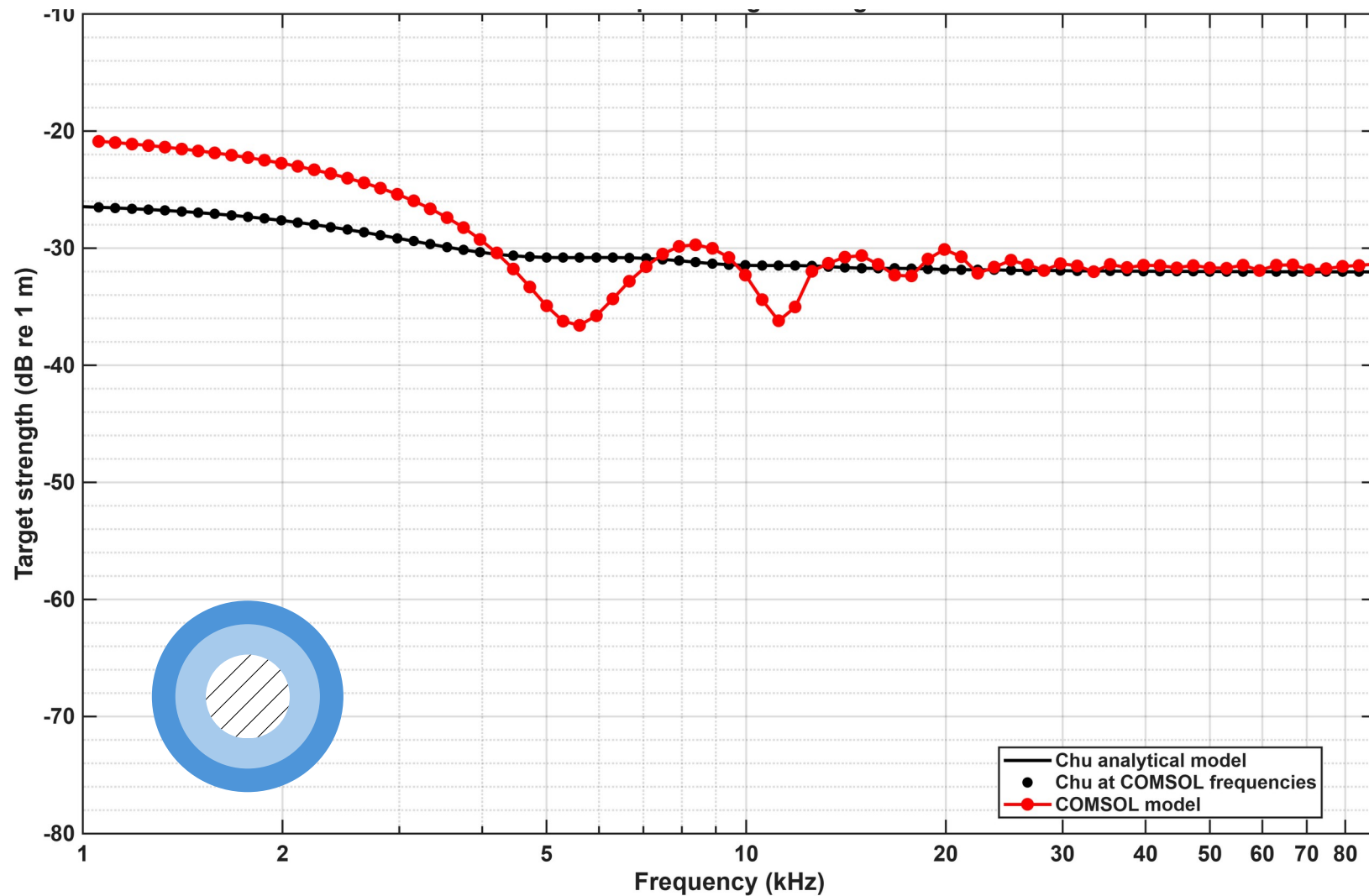


Bone shell
with air cavity

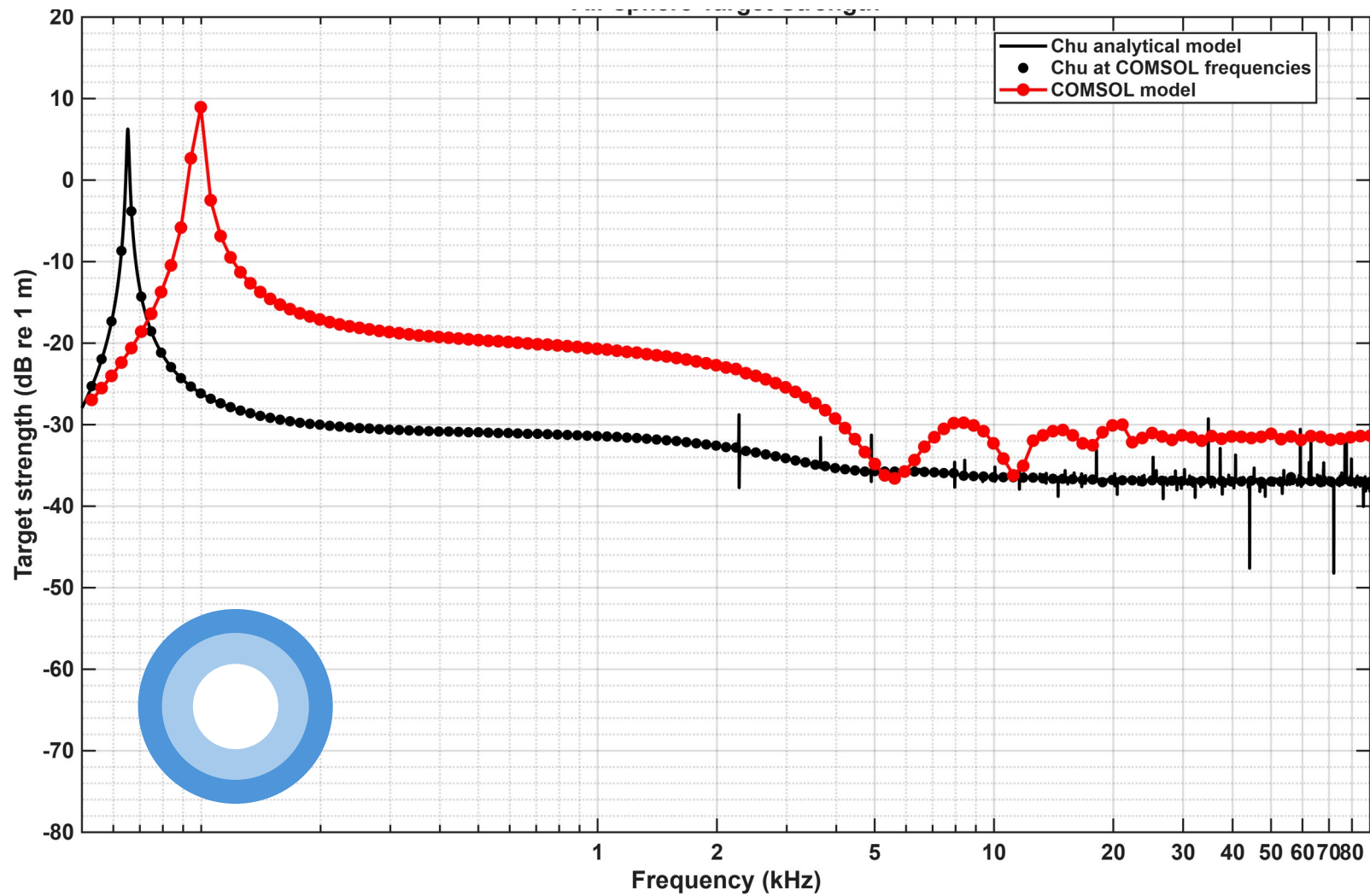
Rigid sphere



Soft sphere



Air Bubble



Air Bubble: Minnaert Resonance

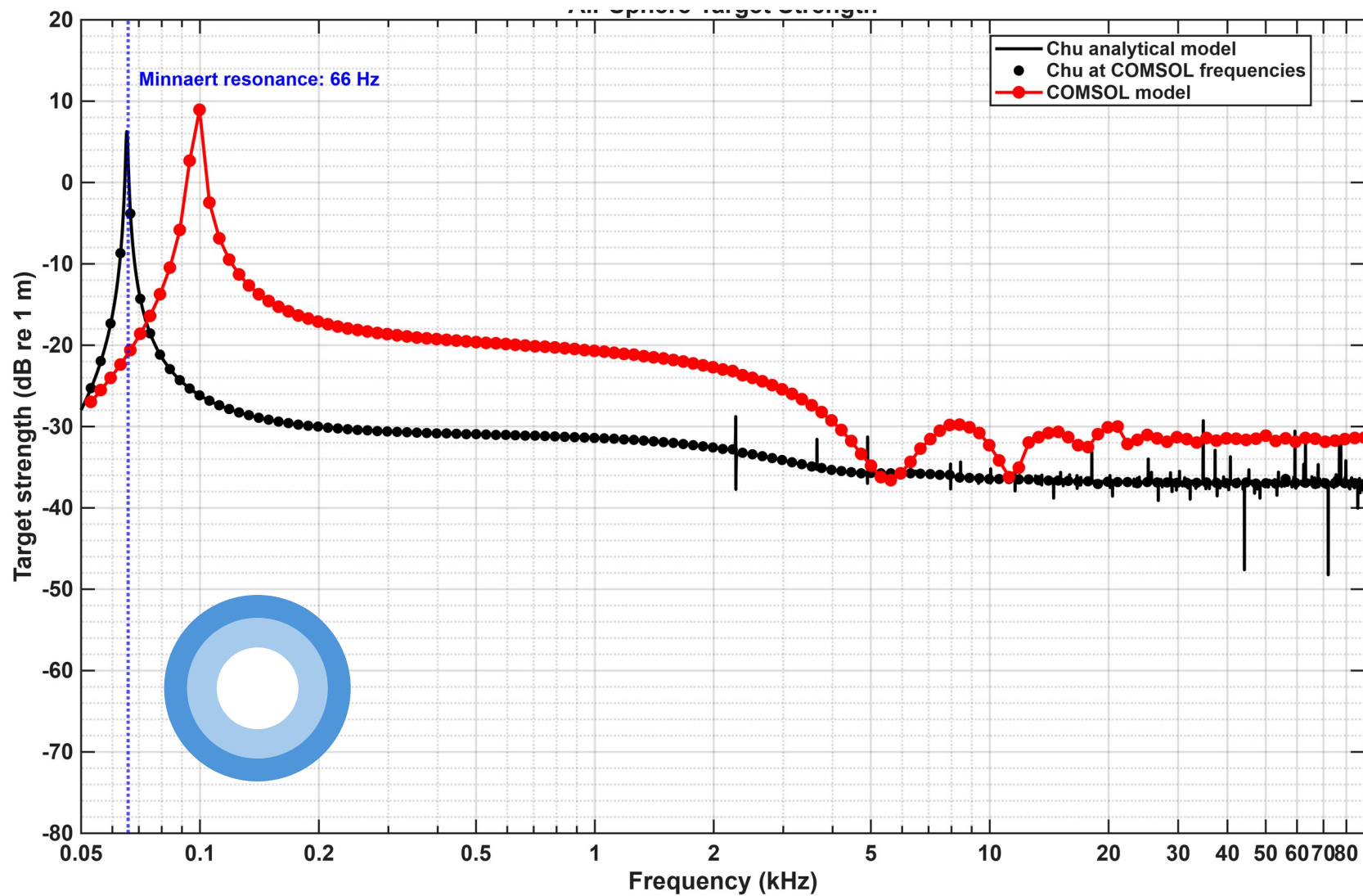


XVI. *On Musical Air-Bubbles and the Sounds of Running Water.* By M. MINNAERT, Sc.D., Heliophysical Institute of the Physical Laboratory, Utrecht*.

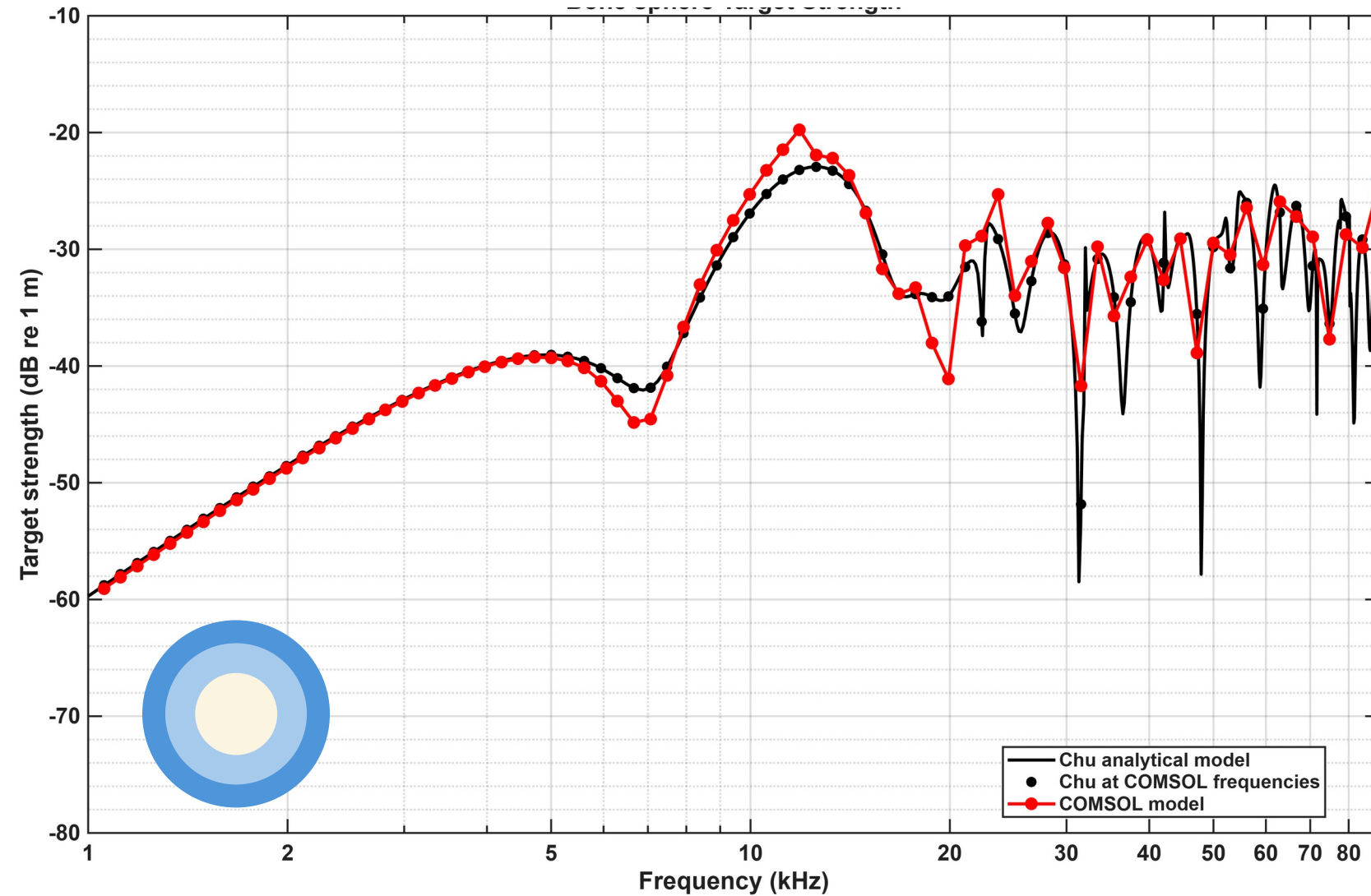
PHYSICISTS have hardly ever investigated the sounds of running water. As a matter of fact we know very little about the murmur of the brook, the roar of the cataract, or the humming of the sea. It is in conformity with the essence of the physical method that we should begin by reducing such acoustical complexes to much simpler elementary phenomena.

$$f_M = \frac{1}{2\pi R} \sqrt{\frac{3\gamma P_0}{\rho_w}}$$

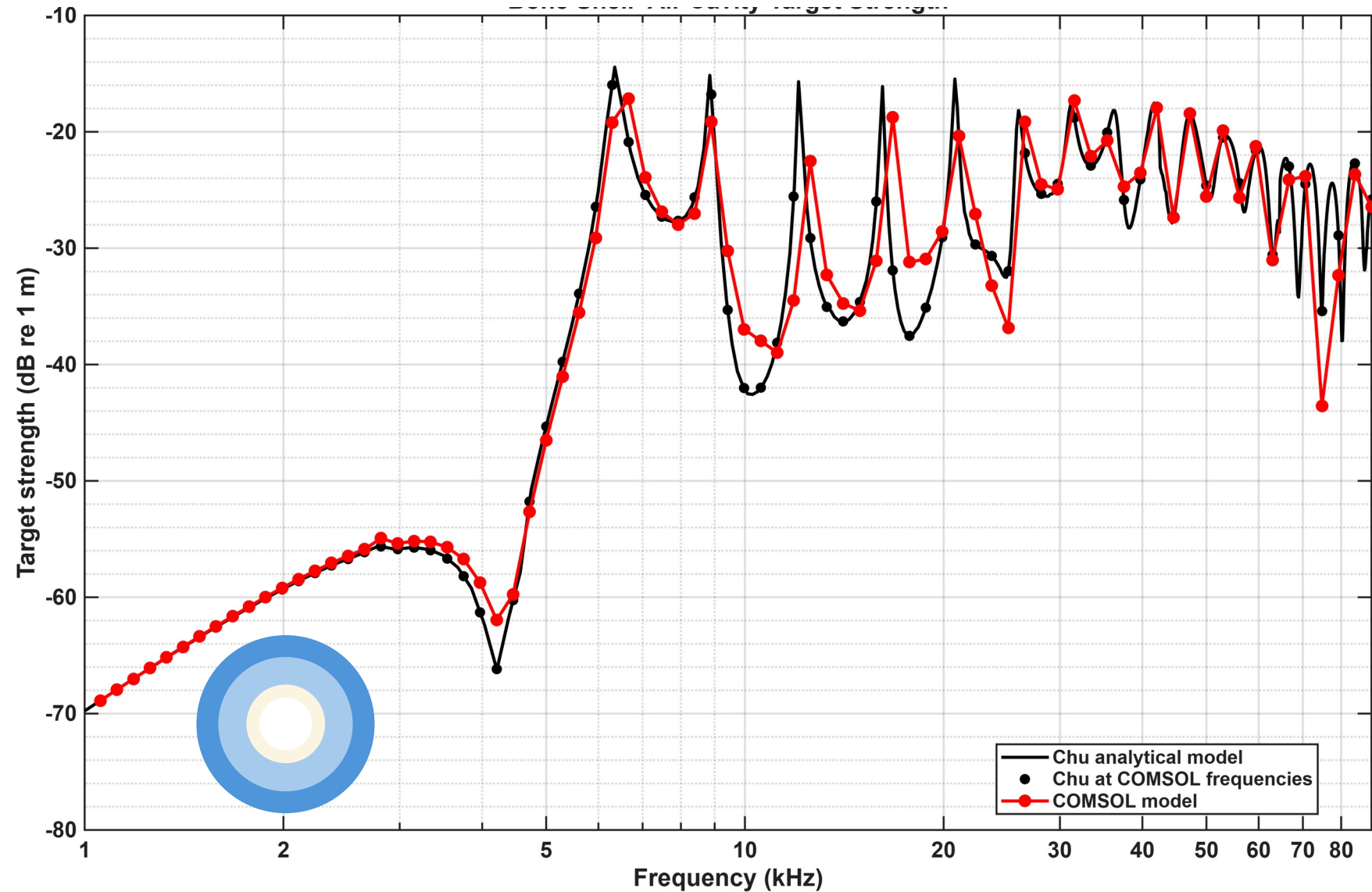
Air Bubble



Bone sphere



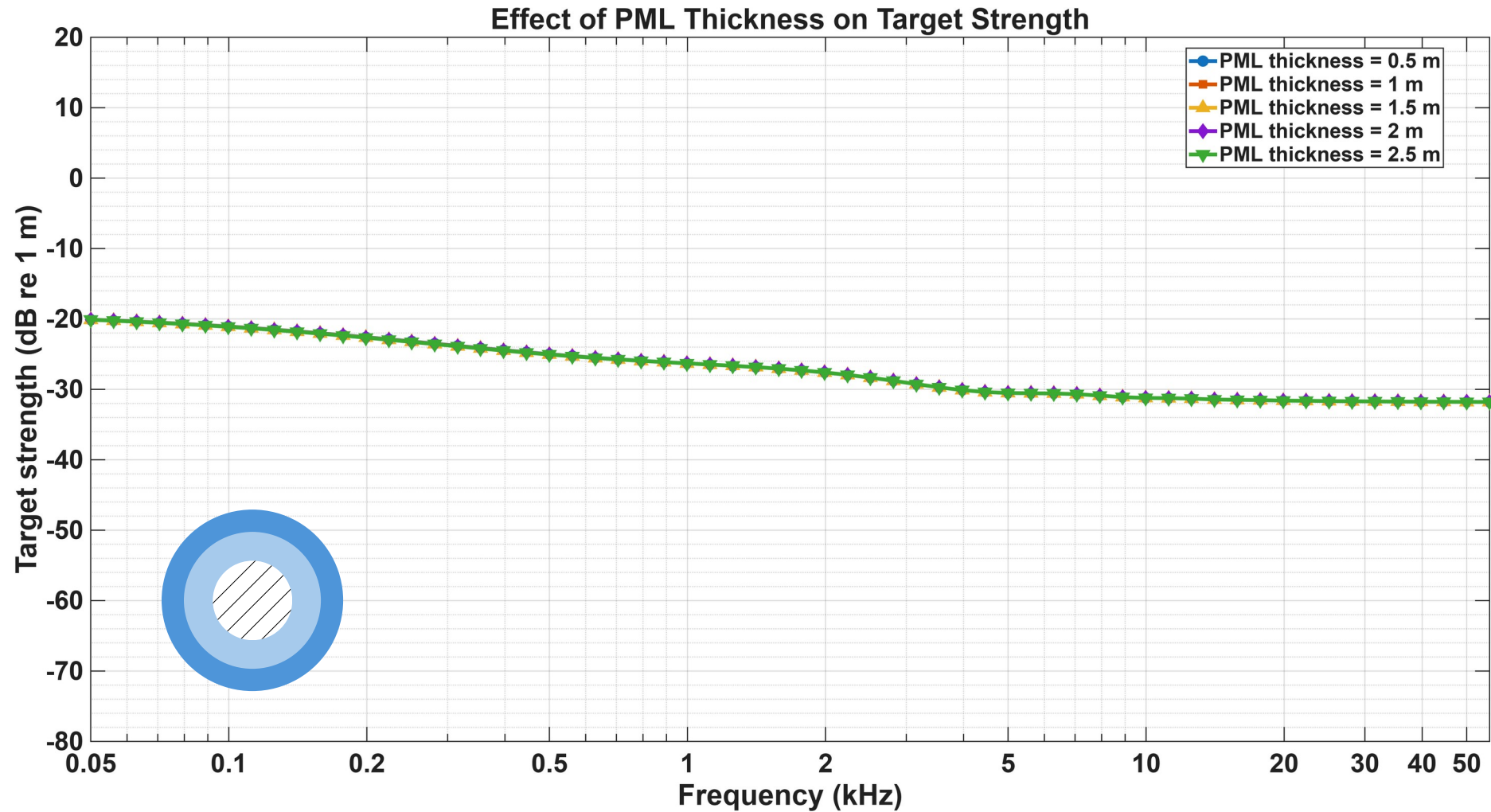
Bone shell with air cavity



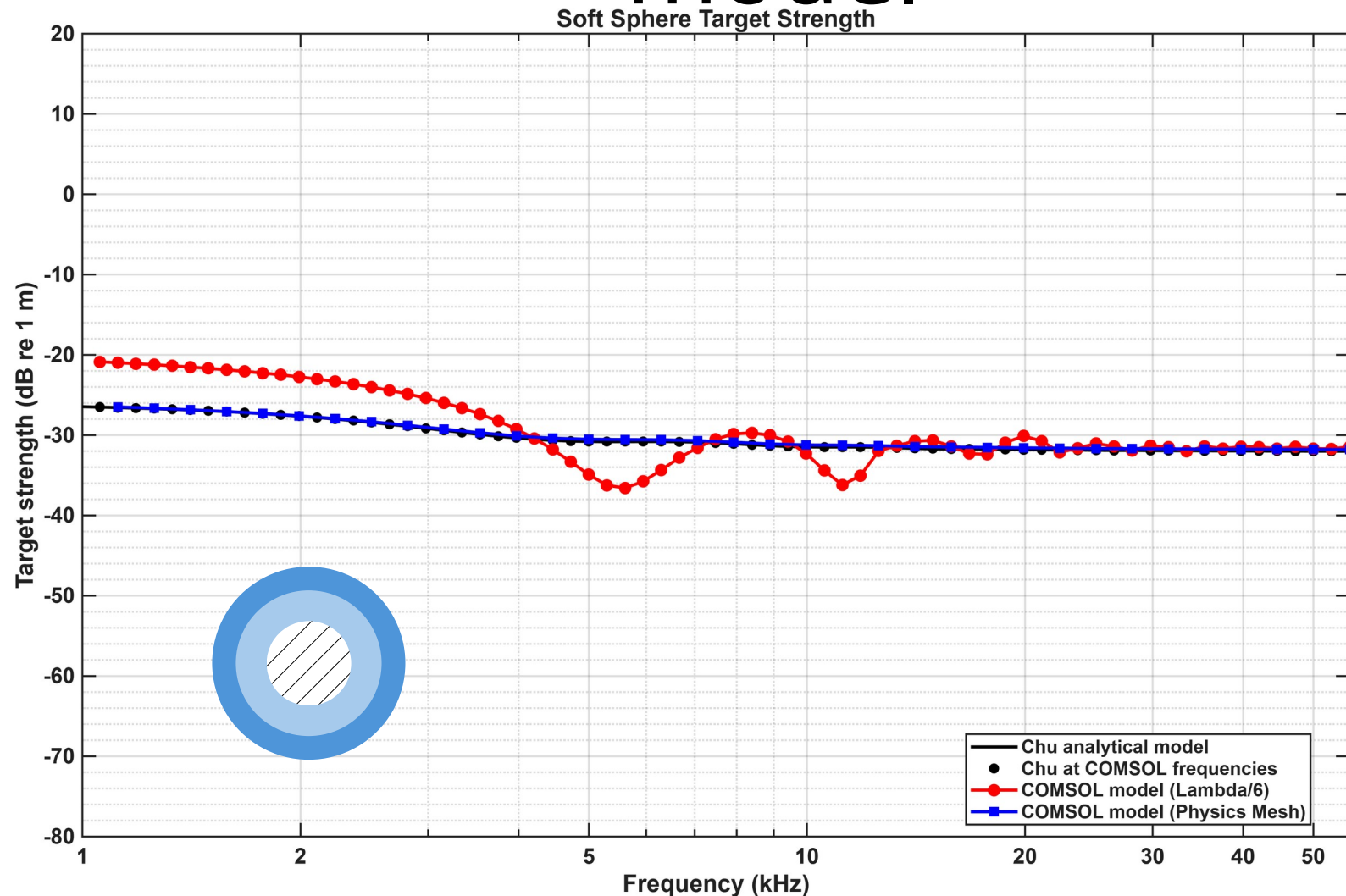
Why this mismatch?

- 1) PML thickness or scaling method
- 2) Inadequate mesh density of scatterer/water domain
- 3) Size of the water domain
- 4) Some stupid mistake

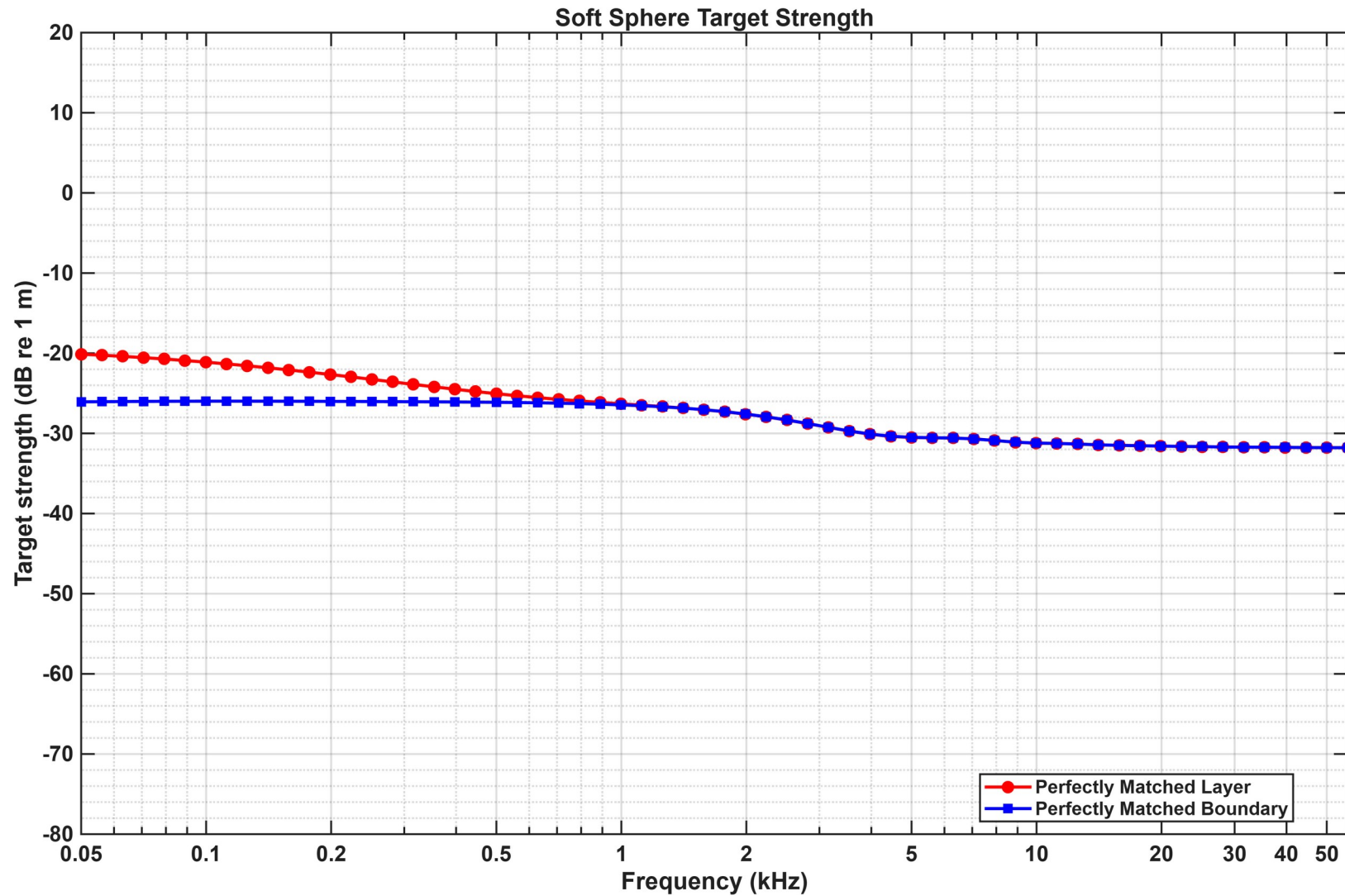
Influence of PML thickness



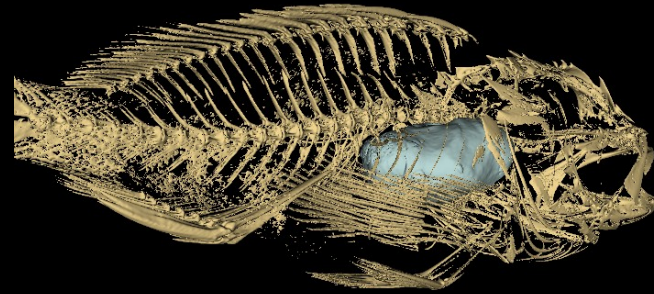
A discovery: physics-based mesh resolves inaccuracies in soft sphere model



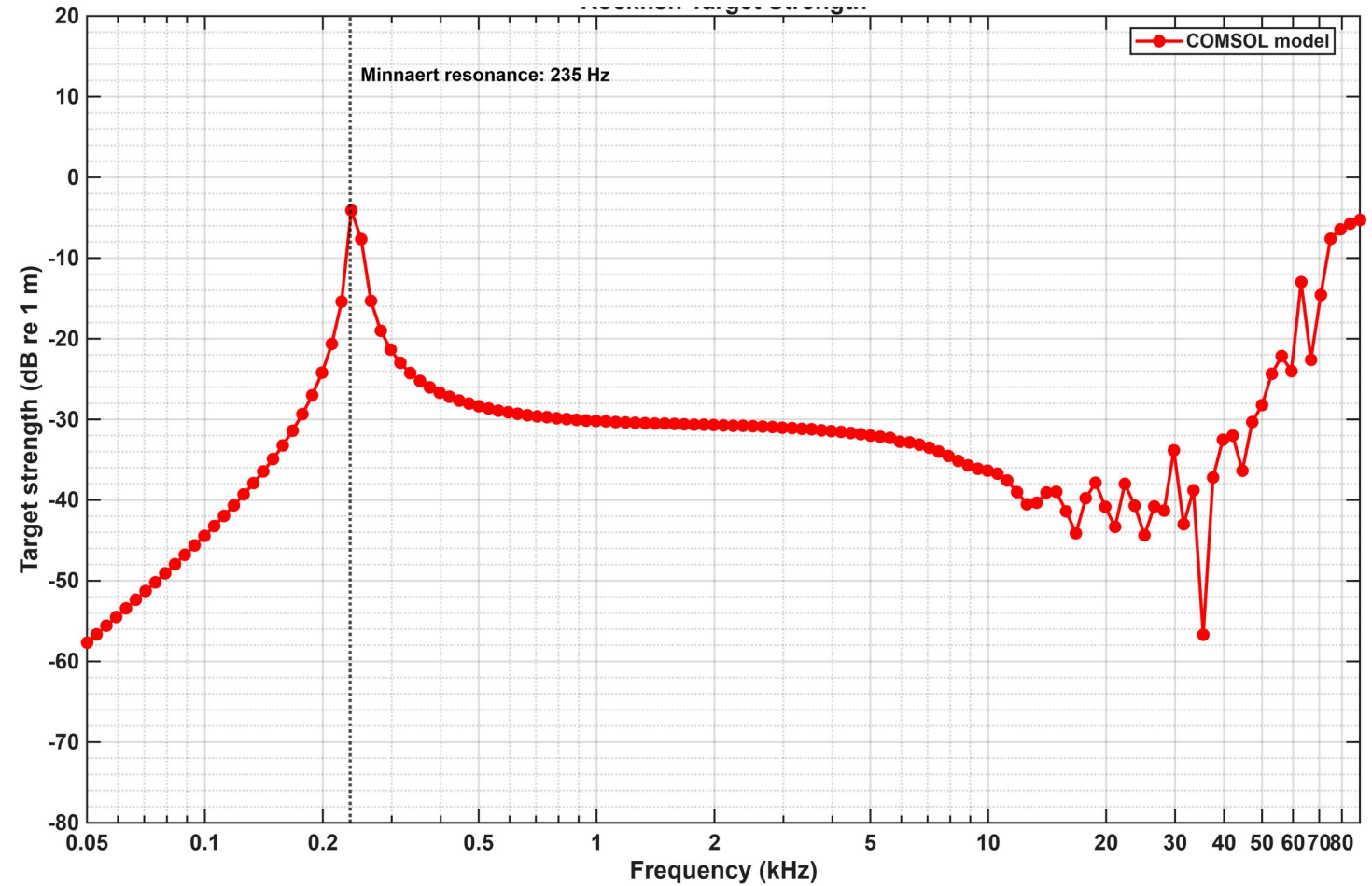
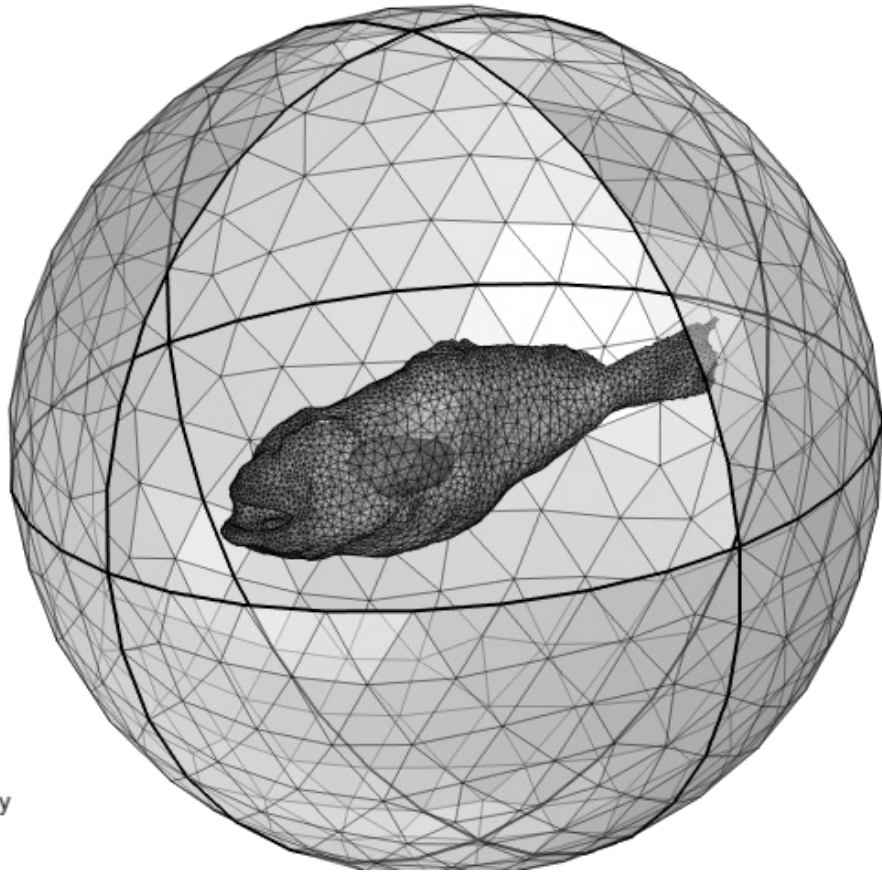
PM Layer vs PM Boundary



Some preliminary experiments with CT data

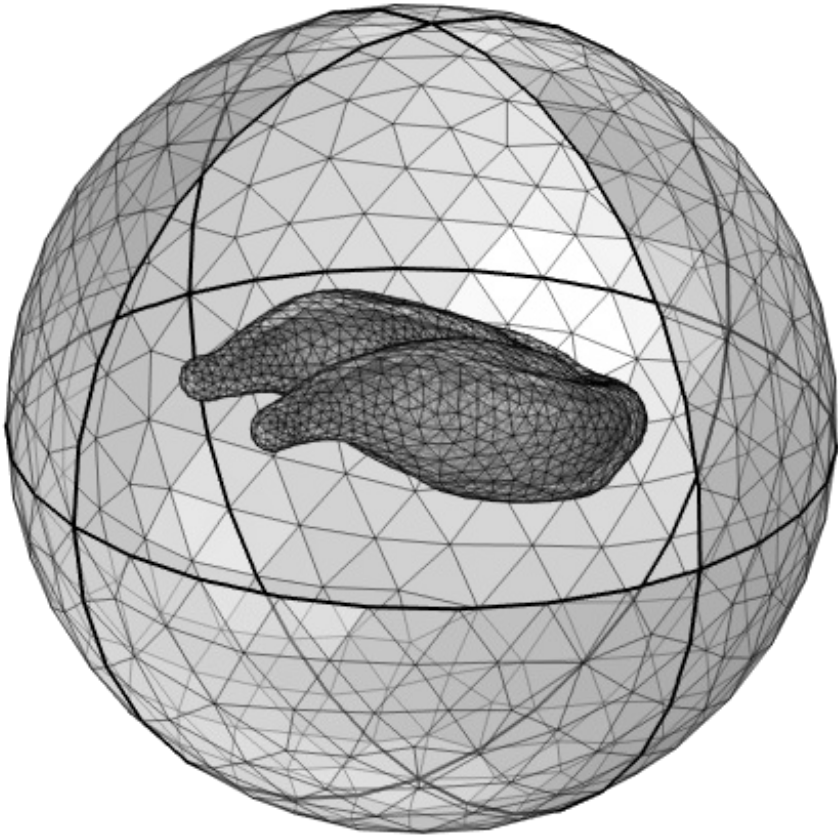


Rockfish scattering function

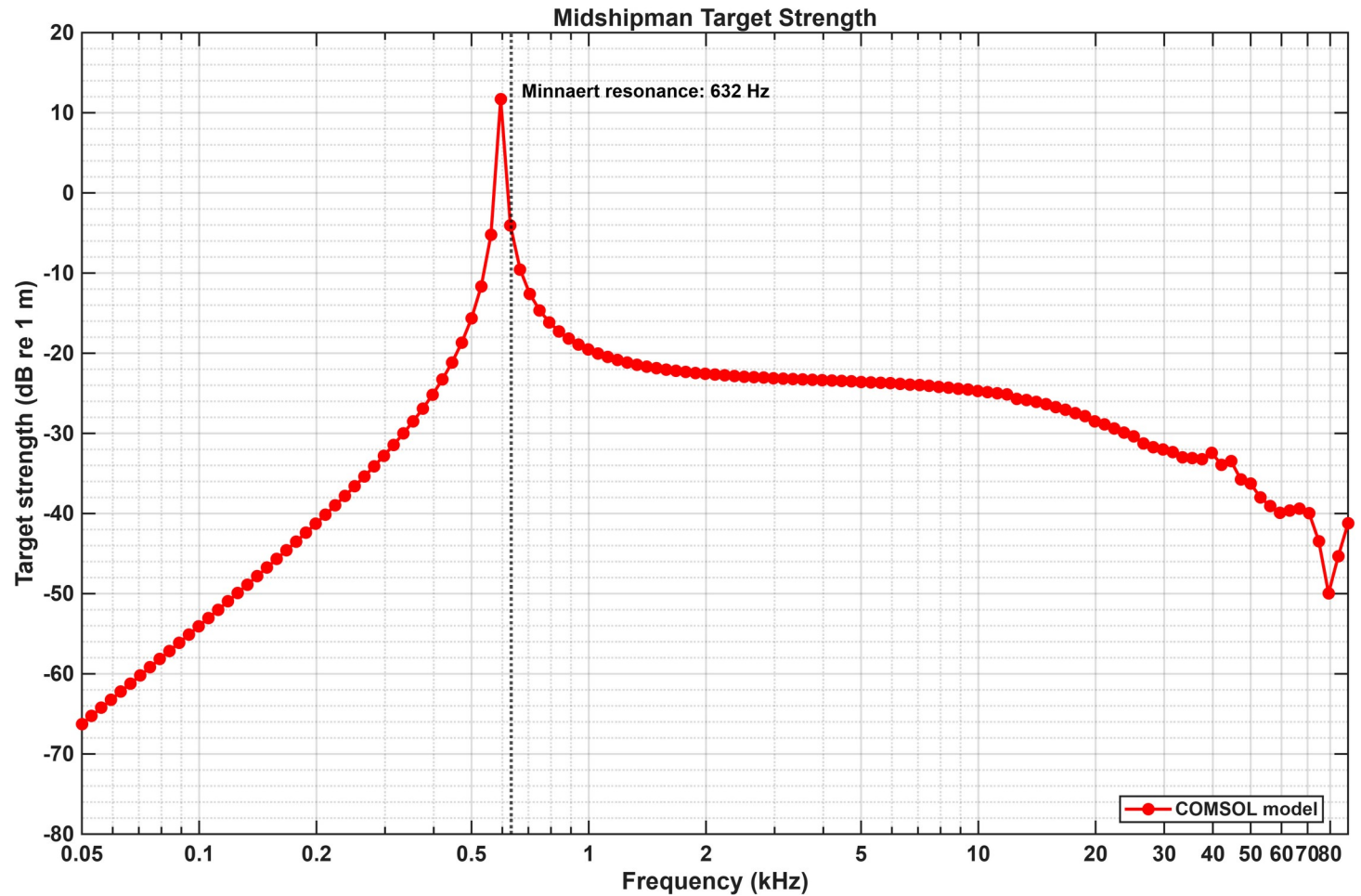


1. Using perfectly matched boundary (PMB)
2. No bony parts assigned yet
3. Volume equivalent radius of SB: 14

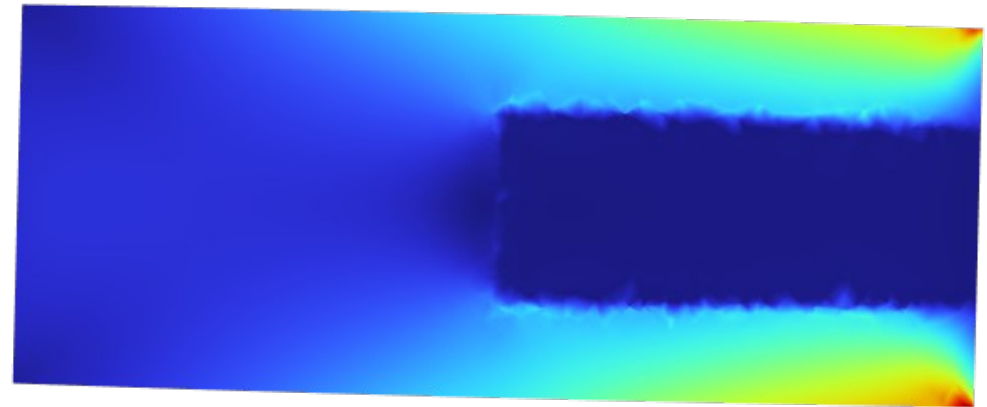
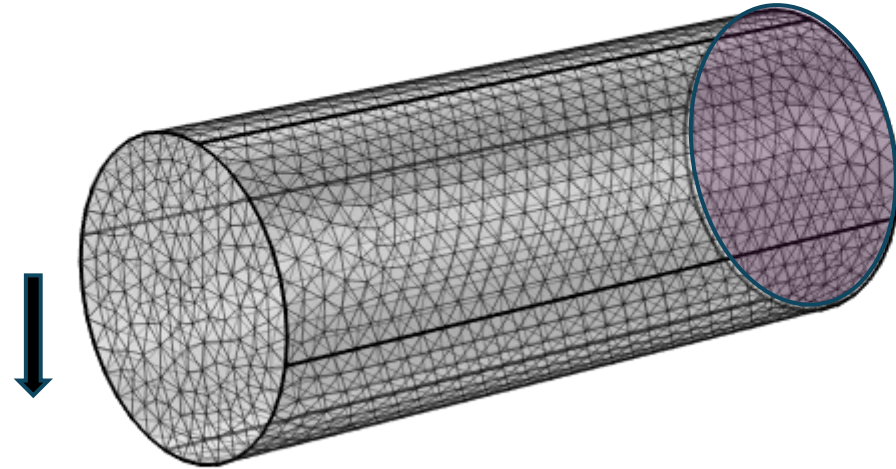
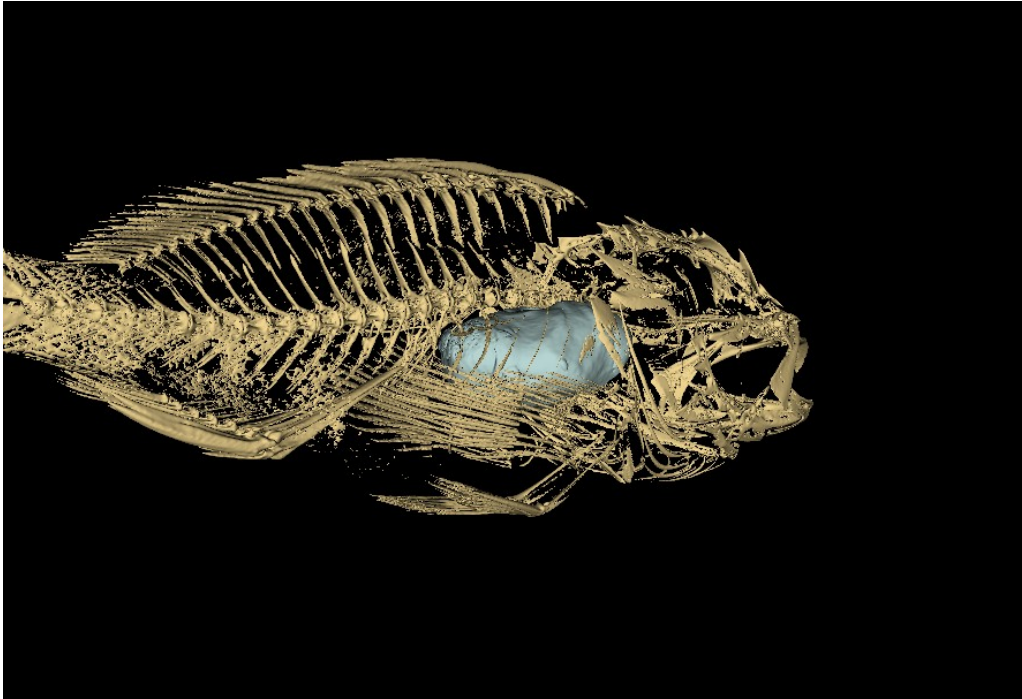
Plainfin midshipman scattering function



1. Using perfectly matched boundary (PMB)
2. Only swim bladder geometry included



In progress: Voxel-based material property assignment



Future directions

Benchmarking

- Run all models with physics-based mesh to check if mismatches are resolved. Examine the differences in mesh sizes between user-specified and physics-based method
- Sanity check with Yoonjeon's elastic sphere model

CT models

- Run more expensive PML based models with rockfish and midshipman
- Develop a method for voxel-based assignment of bone properties